

# AI-Assisted Quantum Simulation & Modelling with Salesforce Agentforce

Shriya Desai<sup>1,\*</sup>, Dr. Ankit Shrivastava<sup>2</sup>, and Dr. Arpita Baronia<sup>3</sup>

Department of CSE, Manipal University Jaipur, Jaipur, India  
ankitshrivastava.ieee@gmail.com

**Abstract.** By combining advanced AI/ML surrogates for quantum modeling with agentic automation for workflow, diagnostics, and user guidance, the project delivers an end-to-end, accessible, and adaptive quantum simulation ecosystem. This not only accelerates science, but also opens quantum modeling to a broader, more diverse audience—pushing the boundaries of both research and education. Most traditional simulation tools offer strong core capabilities but lack workflow guidance, ML integration, and user onboarding. LLM agents provide automation and conversation but can be unreliable and complex for non-experts. Surrogate ML delivers speed but suffers from poor integration and accessibility. This solution takes best practices from these, introducing an accessible, AI-driven assistant that guides, predicts, and interprets throughout the workflow, fully integrating simulation tools and ML surrogates in a modular platform.

**Keywords:** Quantum · Agentic · Simulation · Modelling.

## 1 Introduction

Scientists now treat quantum simulation as a basic instrument as it accelerates discovery. It reveals how complex events unfold inside materials, atoms and molecules, solids, etc. They choose between two separate routes. The first route keeps the work on supercomputers approximating quantum equations with classical chips. The second, more recent route is using digital or analog quantum simulators that leverage quantum computers and quantum devices. Exponential computational cost – Large or highly entangled systems required for correctly simulating quantum systems have resources that increase exponentially with increase in system size. These are intractable even for the largest of supercomputers.

Quantum Hardware resource limits - Present day quantum processors contain only limited qubits. Coherence time is very short, error rates are high and require error correction. As a result, only small systems can be practically simulated.

Parameter tuning - A productive simulation run depends on careful choices - basis sets, cutoff energies, time steps and more. Each choice alters accuracy, speed

or stability. This requires expertise and involves trial-and-error. Miscalculations can lead to wasted computation or meaningless results.

**Diagnostic and interpretive bottlenecks** – Raw simulation data is complex to interpret and remains to be a manual task. Even identifying convergence issues, or diagnosing faulty results is time-consuming and errors are a risk.

**Scalability and accessibility** - Quantum simulation workflows require using command-line and customized scripting, creating accessibility hurdles for new learners and interdisciplinary researchers.

The Higgs boson was first theorised in 1964 by Peter Higgs and was experimentally proven nearly 5 decades later in 2012 at CERN. Though a lot of this time was to engineer the LHC, exploring the vast parameter space, brute force scanning and simulating many high energy interactions, extracting rare signals from large datasets took decades too. This shows how computationally and experimentally intensive modern physics is.

Recent advancements in AI/ML can help alleviate these bottlenecks, particularly surrogate models based on Graph Neural Networks can potentially help approximate expensive simulations, guide parameter exploration, and accelerate discovery pipelines, especially by incorporating agentic AI into the workflows.

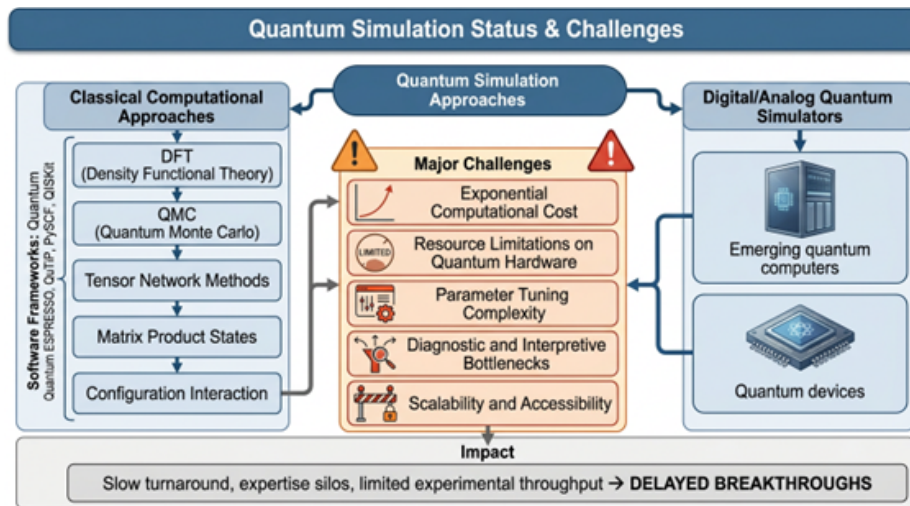


Fig. 1. Quantum Simulation

## 2 Related Work

In quantum mechanics, the wavefunction tells us the complete characteristics and behaviour of a quantum system. For a large system containing many particles, this wavefunction is very difficult to computationally calculate.

This is where NNQS comes in. Neural Network Quantum State is an ML surrogate that becomes the model of the quantum state. Instead of calculating the wavefunction, our NN learns to approximate it.

The first paper that introduced this idea is the Carleo and Troyer (2017): "Solving the Quantum Many-Body Problem with Artificial Neural Networks". Another paper describing how Agentic AI helps optimize scientific workflows is the "Müller et al. (2020): "Revolution of Scientific Workflows in the Agentic AI Era". It shows how Agentic AI reduces manual effort and hence accelerates discovery.

This model applies both NN surrogates and agentic workflows directly to practical quantum simulation tasks (parameter selection, interpreting outputs, estimating results).

## 3 Solution

This paper provides a way to revolutionize quantum simulation workflows by integrating AI/ML in two key areas:

### 1. Surrogate Modelling of Quantum Simulations

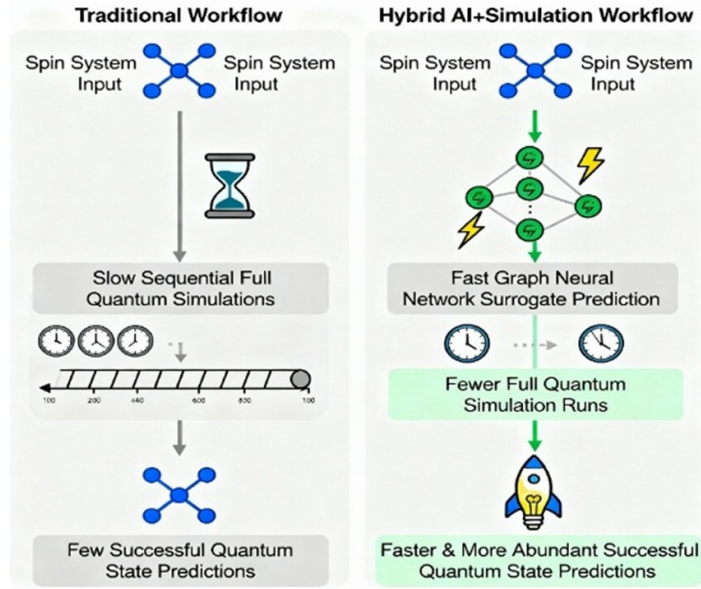
Rather than running calculations and simulations for every quantum system or parameter scenario, which may lead to wasted computations, ML surrogates based on neural architectures like Graph Neural Networks (GNN) can be leveraged. Surrogate models can rapidly predict quantum properties and solutions in new scenarios and is trained on pre-existing sets of experimental or simulated results. This can help in screening candidates (e.g., for materials or molecules) by allowing workflows to only prioritize simulations for promising subsets.

ML Selection Rationale:

Each quantum system (e.g., molecule, spin chain) encoded as a graph:

- Nodes: Atoms (features: element type, charge, quantum state)
- Edges: Bonds/interactions (features: bond type, distance, coupling constant)

While simpler models (Linear, MLP, SVM) provide rapid prototyping and interpretability, they are unable to capture the highly interconnected, non-linear relationships present in quantum systems, especially molecules, crystals, or spin networks, which are naturally represented as graphs. GNNs can process node and edge features, update representations via message passing, and perform efficient pooling/readout for graph-level prediction, hence being considered superior in this scientific domain.



**Fig. 2.** Workflow Comparison

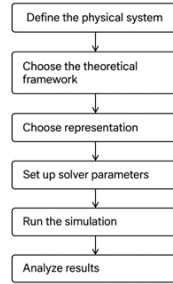
## 2. Agentic AI for end-to-end Quantum Simulation

Achieving a truly transformative quantum simulation workflow requires integration of parameter selection and surrogate outcomes that decide for or against simulation into a coherent, user-friendly platform. This paper suggests the orchestration of an agentic AI, specifically Salesforce Agentforce, to build a conversational, automated agent to make the workflow accessible. Here, Salesforce Agentforce acts as the ‘brain’ of the assistant, orchestrating user interactions, ML surrogate calls whose outcomes determine whether backend simulation tasks are required and interpreting the results of these processes.

Key Capabilities:

- Users start simulations in their respective quantum problem space
- The agent triggers ML surrogates with multiple sets of simulation parameters
- ML surrogates are invoked for instant predictions, rapidly narrowing down feasible or interesting parameter sets and flagging scenarios for detailed quantum calculation.
- Once the optimal scenario is identified, the agent triggers full-physics quantum simulations via tools such as QuTiP or Qiskit, tracking execution and collecting comprehensive results.
- Post-simulation, the agent parses and interprets results.

Quantum Simulation Workflow



**Fig. 3.** Simulation Workflow

## 4 Objectives

- Demonstrating power of AI/ML in optimization of quantum simulations by reducing computational cost.
- Integrating an agentic AI to convert scientific workflows into conversational interactions for scientists.
- Allowing students or non-experts to run quantum simulations without in depth numerical or coding knowledge.
- Creating a blueprint for future agentic AI assisted scientific workflows.

## 5 Methodology

The proposed methodology follows a controlled and iterative workflow spanning agentic orchestration, machine-learning-based surrogate modeling, and traditional quantum simulation.

The objective is to augment the standard scientific quantum simulators with a rich decisioning and optimizing model that provides a refined and more higher probability parameter and configuration subset that helps reduce outliers and improves speed and accuracy of processing and also provides a structure to analyse and interpret outcomes.

At a high level, this system starts with user-defined scientific objective that fed to an AI agent which then analyses and identifies the relevant simulation pathway, and routes the task through either a fast surrogate estimate or a higher-fidelity quantum solver depending on the diagnostic outcome.

1. Agentic intake The process begins with the scientific objective expressed in natural language. The agentic layer translates this into a structured workflow by identifying the simulation type, the expected outputs, and the key parameter groups that must be evaluated.

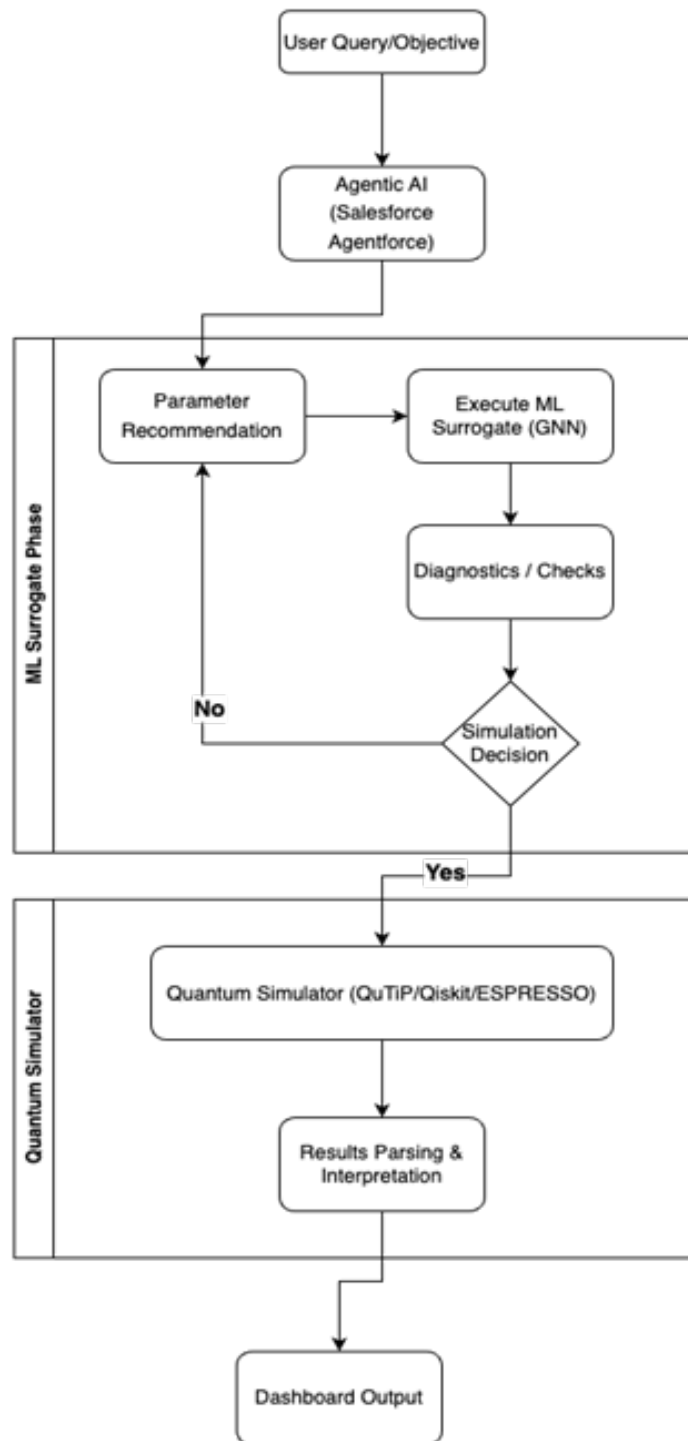


Fig. 4. End-to-End Workflow

2. Surrogate-guided parameter refinement The next step is using a surrogate model for parameter refinement. The surrogate model is used for evaluating parameter choices and combinations to identify the most probable subset of choices that can then be fed into the quantum simulation model for further scientific analysis and decisioning. This step helps reduce the problem space for the subsequent stimulation activities thus providing critical time saving and faster outcomes compared to traditional brute-force simulation approaches where the entire problem space is evaluated.

3. Diagnostics and decision control The next step is diagnostics and decisioning that help identify inconsistent parameter configurations and determination of whether to rerun the surrogate model to generate a new set of options or invoke the quantum simulator with the configuration options generated from the surrogate model. This step provides a control mechanism in the workflow before the expensive quantum simulation is initiated.

4. Quantum simulation and authoritative validation The quantum simulation is the final evaluation phase that will provide the final outcomes and scientific results for the provided problem statement. The quantum simulator is the trusted backend that validates the surrogate-guided selection. This helps preserve the rigor of traditional quantum computation. This hybrid structure of a surrogate model led quantum simulation leverages the computational advancements in machine learning to accelerate exploration, while continuing to base scientific analysis on domain-specific quantum simulator for scientific outcomes.

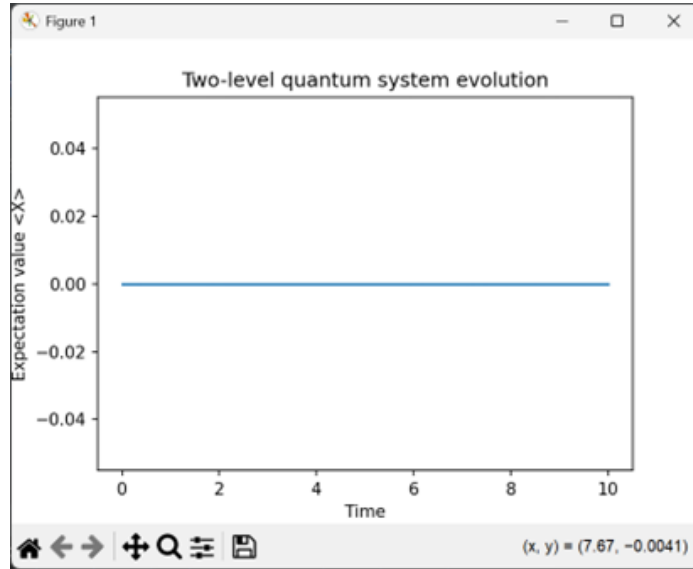
5. Result interpretation and reporting The final phase of the workflow is output interpretation. The simulation results are parsed and summarized. The system not only provides the raw numerical values, but also provides an analysis of the results and how they map to original problem statement. The dashboard then presents the final output allowing the workflow to serve both researchers and learners. This reporting layer thus helps turn a complex scientific computation into a practical and reusable research aid.

**Workflow Interpretation** The diagram therefore represents a two-stage computational strategy wrapped inside an agentic control framework. The initial stages leverage AI for planning, parameter recommendation, and surrogate inference, while the later stages provide the scientific rigor using quantum simulation and result presentation. Conceptually, this separation reflects the project’s philosophy: first narrow the problem intelligently, then apply the scientific approach to problem solving where it matters most.

## 6 Results

The workflow design successfully connects user intent, surrogate prediction, diagnostics, and quantum simulation into one clear pipeline.

The surrogate stage helps narrow down simulation scenarios before costly computation is used.



**Fig. 5.** Simulation

The diagnostic step adds a useful checkpoint for catching poor parameter choices early.

The overall structure shows that AI can improve both speed and usability in quantum simulation research.

## 7 Conclusion

Surrogate models help in parameter screening by providing predictions based on given thresholds.

Agentic AI simplifies complex scientific workflows by guiding surrogate predictions and simulations, and interpreting the results, instead of everything being done manually

## 8 Future Work

### Research Directions:

Beyond the current approach, there are multiple further research avenues - both expanding the system intelligence implementing active learning loops and expanding the domains of use; as well as industrialized the current solution into a research-platform grade solution that can be scaled across multi-users and use cases.

Active Learning Loops using multi-scale simulation and uncertainty quantification : Implement ML models that suggest optimal parameter sets for simulation based on uncertainty quantification and exploration-exploitation balance. Combine fast ML surrogates with high-fidelity DFT/Quantum ESPRESSO runs, automatically selecting the right fidelity for each scenario. Add Bayesian neural networks or ensemble methods to surrogate models for reliable confidence intervals on predictions.

Domain Expansion: Adapt the agentic workflow to other scientific domains (materials discovery, protein folding, quantum chemistry reaction prediction).

#### **Production and Scalability:**

Multi-User Support: Add user authentication, simulation queuing, and result persistence with database integration.

Performance Benchmarks: Quantify speedup vs traditional workflows through controlled experiments across molecular libraries.

## **9 Acknowledgements**

I would like to thank my mentor Dr. Ankit Shrivastava for his meaningful assistance, tireless patience and guidance. I would like to extend my appreciation towards Dr. Arpita Baronia for her continued support. I would also like to thank my family for their moral support and guidance.

## **10 References**

Bensoussan, A., Chachkarova, E., Even-Mendoza, K., Fortz, S., and Lenihan, C. (2025). AccelerQ: Accelerating quantum eigensolvers with machine learning on quantum simulators. *Proceedings of the ACM on Programming Languages*, 9(OOPSLA2), Article 354, 1–31. <https://doi.org/10.1145/3763132>

Zou, Y., Cheng, A. H., Aldossary, A., Bai, J., Leong, S. X., Campos-Gonzalez-Angulo, J. A., ... and Aspuru-Guzik, A. (2025). El Agente: An autonomous agent for quantum chemistry. *Matter*, 8(7).

<https://www.spinquanta.com/news-detail/spinq-quantum-artificial-intelligence-lab-research-project-overview>

Carleo, G., and Troyer, M. (2017). Solving the quantum many-body problem with artificial neural networks. *Science*, 355(6325), 602–606. <https://doi.org/10.1126/science.aag2302>

Shin, W., Souza, R., Rosendo, D., Suter, F., Wang, F., Balaprakash, P., and da Silva, R. F. (2025). The (R)evolution of scientific workflows in the agentic AI era: Towards autonomous science. In *Proceedings of the 20th Workshop on Workflows in Support of Large-Scale Science (WORKS '25)* (pp. 2305–2316). Association for Computing Machinery. <https://doi.org/10.1145/3731599.3767580>